My Project

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Chapter 1

Hierarchical Index

1.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

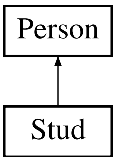
Chapter 4

Class Documentation

4.1 Person Class Reference

```
#include <header.h>
```

Inheritance diagram for Person:



Public Member Functions

- Person ()
- Person (string name, string last_name)
- string get_name () const
- string get_last_name () const
- virtual void about () const =0
- virtual ~Person ()

Protected Attributes

- string name_
- string last_name_

4.1.1 Constructor & Destructor Documentation

4.1.1.1 Person() [1/2]

```
Person::Person ()  [inline]
```

4.1.1.2 Person() [2/2]

```
Person::Person (
            string name,
            string last_name)  [inline]
```

4.1.1.3 ∼Person()

```
virtual Person::∼Person ()  [inline], [virtual]
```

4.1.2 Member Function Documentation

4.1.2.1 about()

```
virtual void Person::about () const  [pure virtual]
```

Implemented in Stud.

4.1.2.2 get_last_name()

```
string Person::get_last_name () const  [inline]
```

4.1.2.3 get_name()

```
string Person::get_name () const  [inline]
```

4.1.3 Member Data Documentation

4.1.3.1 last_name_

```
string Person::last_name_  [protected]
```

4.1.3.2 name_

```
string Person::name_  [protected]
```

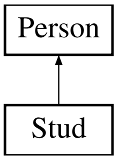
The documentation for this class was generated from the following file:

- header.h

4.2 Stud Class Reference

```
#include <header.h>
```

Inheritance diagram for Stud:



Public Member Functions

- Stud ()
- Stud (string name, string last_name)
- Stud (std::stringstream &is, int number_of_homework_marks)
- Stud (const Stud &other)
- Stud (Stud &&other) noexcept
- Stud & operator= (const Stud &other)
- Stud & operator= (Stud &&other) noexcept
- int get_exam_mark () const
- void set_exam_mark (int exam_mark)
- void set_homework_marks (int Homework_mark)
- void clean_homework_marks ()
- void set_final_mark (double final_mark)
- double get_final_mark () const
- void set_second_final_mark (double second_final_mark)
- double get_second_final_mark () const
- double Get_average_for_homework_mark ()
- double Get_mediana_for_homework_mark ()
- void generate_marks ()
- void generate_name ()
- void about () const override
- ∼Stud ()

Public Member Functions inherited from Person

- Person ()
- Person (string name, string last_name)
- string get_name () const
- string get_last_name () const
- virtual ∼Person ()

Friends

- std::ostream & operator<< (std::ostream &out, const Stud &student)
- std::istream & operator>> (std::istream &in, Stud &student)

Additional Inherited Members

Protected Attributes inherited from Person

- string name_
- string last_name_

4.2.1 Constructor & Destructor Documentation

4.2.1.1 Stud() [1/5]

```
Stud::Stud ()  [inline]
```

4.2.1.2 Stud() [2/5]

```
Stud::Stud (
            string name,
            string last_name)  [inline]
```

4.2.1.3 Stud() [3/5]

```
Stud::Stud (
            std::stringstream & is,
            int number_of_homework_marks)
```

4.2.1.4 Stud() [4/5]

```
Stud::Stud (
            const Stud & other)  [inline]
```

4.2.1.5 Stud() [5/5]

```
Stud::Stud (
            Stud && other)  [inline], [noexcept]
```

4.2.1.6 ∼Stud()

```
Stud::∼Stud ()
```

4.2.2 Member Function Documentation

4.2.2.1 about()

```
void Stud::about () const  [inline], [override], [virtual]
```

Implements Person.

4.2.2.2 clean_homework_marks()

```
void Stud::clean_homework_marks ( ) [inline]
```

4.2.2.3 generate_marks()

```
void Stud::generate_marks ( )
```

4.2.2.4 generate_name()

```
void Stud::generate_name ( )
```

4.2.2.5 Get_average_for_homework_mark()

```
double Stud::Get_average_for_homework_mark ( )
```

4.2.2.6 get_exam_mark()

```
int Stud::get_exam_mark ( ) const [inline]
```

4.2.2.7 get_final_mark()

```
double Stud::get_final_mark ( ) const [inline]
```

4.2.2.8 Get_mediana_for_homework_mark()

```
double Stud::Get_mediana_for_homework_mark ( )
```

4.2.2.9 get_second_final_mark()

```
double Stud::get_second_final_mark ( ) const [inline]
```

4.2.2.10 operator=() [1/2]

```
Stud & Stud::operator= (
            const Stud & other)
```

4.2.2.11 operator=() [2/2]

```
Stud & Stud::operator= (
            Stud && other) [noexcept]
```

4.2.2.12 set_exam_mark()

```
void Stud::set_exam_mark (
            int exam_mark)  [inline]
```

4.2.2.13 set_final_mark()

```
void Stud::set_final_mark (
            double final_mark)  [inline]
```

4.2.2.14 set_homework_marks()

```
void Stud::set_homework_marks (
            int Homework_mark)  [inline]
```

4.2.2.15 set_second_final_mark()

```
void Stud::set_second_final_mark (
            double second_final_mark)  [inline]
```

4.2.3 Friends And Related Symbol Documentation

4.2.3.1 operator<<

```
std::ostream & operator<< (
            std::ostream & out,
            const Stud & student)  [friend]
```

4.2.3.2 operator>>

```
std::istream & operator>> (
            std::istream & in,
            Stud & student)  [friend]
```

The documentation for this class was generated from the following files:

- header.h
- functions.cpp

Chapter 5

File Documentation

5.1 functions.cpp File Reference

```
#include "header.h"
```

Functions

- void Get_final_mark (vector< Stud > &grupe, bool for_average_homework_mark, bool for_both_homework_mark)
- void Print_final_mark (vector< Stud > &grupe, bool for_average_homework_mark, bool for_both_homework_mark, bool print_results_in_terminal)
- int Get_size_for_string_printing (vector< Stud > &grupe)
- void generate_marks (vector< int > &Marks, int number_of_marks)
- void Sort_students (vector< Stud > &grupe, string parametr)
- bool Generate_file_with_students (int number_of_students, int number_of_marks, string filename)
- void Divide_for_two_grupse (vector< Stud > &grupe, vector< Stud > &best_grupe, vector< Stud > &worst_grupe)
- void Divide_for_two_grupse (vector< Stud > &grupe, vector< Stud > &worst_grupe)
- void Divide_for_two_grupse_v3 (vector< Stud > &grupe, vector< Stud > &best_grupe, vector< Stud > &worst_grupe)
- void Enter_students_using_txt_file_bufer_P (vector< Stud > &grupe)
- std::ostream & operator<< (std::ostream &out, const Stud &student)
- std::istream & operator>> (std::istream &in, Stud &student)

Variables

- double time_of_generating_file = 0
- double time_of_reading_file = 0
- double time_of_dividing = 0
- double time_of_sorting = 0
- double time_of_culculating = 0
- double time_of_writing_files = 0

5.1.1 Function Documentation

5.1.1.1 Divide_for_two_grupse() [1/2]

```
void Divide_for_two_grupse (
            vector< Stud > & grupe,
            vector< Stud > & best_grupe,
            vector< Stud > & worst_grupe)
```

5.1.1.2 Divide_for_two_grupse() [2/2]

```
void Divide_for_two_grupse (
            vector< Stud > & grupe,
            vector< Stud > & worst_grupe)
```

5.1.1.3 Divide_for_two_grupse_v3()

```
void Divide_for_two_grupse_v3 (
            vector< Stud > & grupe,
            vector< Stud > & best_grupe,
            vector< Stud > & worst_grupe)
```

5.1.1.4 Enter_students_using_txt_file_bufer_P()

```
void Enter_students_using_txt_file_bufer_P (
            vector< Stud > & grupe)
```

5.1.1.5 Generate_file_with_students()

```
bool Generate_file_with_students (
            int number_of_students,
            int number_of_marks,
            string filename)
```

5.1.1.6 generate_marks()

```
void generate_marks (
            vector< int > & Marks,
            int number_of_marks)
```

5.1.1.7 Get_final_mark()

```
void Get_final_mark (
            vector< Stud > & grupe,
            bool for_average_homework_mark,
            bool for_both_homework_mark)
```

5.1.1.8 Get_size_for_string_printing()

```
int Get_size_for_string_printing (
            vector< Stud > & grupe)
```

5.1.1.9 operator<<()

```
std::ostream & operator<< (
            std::ostream & out,
            const Stud & student)
```

5.1.1.10 operator>>()

```
std::istream & operator>> (
            std::istream & in,
            Stud & student)
```

5.1.1.11 Print_final_mark()

```
void Print_final_mark (
            vector< Stud > & grupe,
            bool for_average_homework_mark,
            bool for_both_homework_mark,
            bool print_results_in_terminal)
```

5.1.1.12 Sort_students()

```
void Sort_students (
            vector< Stud > & grupe,
            string parametr)
```

5.1.2 Variable Documentation

5.1.2.1 time_of_culculating

```
double time_of_culculating = 0
```

5.1.2.2 time_of_dividing

```
double time_of_dividing = 0
```

5.1.2.3 time_of_generating_file

```
double time_of_generating_file = 0
```

5.1.2.4 time_of_reading_file

```
double time_of_reading_file = 0
```

5.1.2.5 time_of_sorting

```
double time_of_sorting = 0
```

5.1.2.6 time_of_writing_files

```
double time_of_writing_files = 0
```

5.2 header.h File Reference

```
#include <iostream>
#include <iomanip>
#include <string>
#include <vector>
#include <random>
#include <fstream>
#include <sstream>
#include <chrono>
#include <algorithm>
```

Classes

- class Person
- class Stud

Functions

- void Get_final_mark (vector< Stud > &grupe, bool for_average_homework_mark, bool for_both_homework_mark)
- void Print_final_mark (vector< Stud > &grupe, bool for_average_homework_mark, bool for_both_homework_mark, bool print_results_in_terminal)
- int Get_size_for_string_printing (vector< Stud > &grupe)
- void generate_marks (vector< int > &Marks, int number_of_marks)
- void Sort_students (vector< Stud > &grupe, string parametr)
- bool Generate_file_with_students (int number_of_students, int number_of_marks, string filename)
- void Divide_for_two_grupse (vector< Stud > &grupe, vector< Stud > &best_grupe, vector< Stud > &worst_grupe)
- void Divide_for_two_grupse (vector< Stud > &grupe, vector< Stud > &worst_grupe)
- void Divide_for_two_grupse_v3 (vector< Stud > &grupe, vector< Stud > &best_grupe, vector< Stud > &worst_grupe)
- void Enter_students_using_txt_file_bufer_P (vector< Stud > &grupe)

Variables

- double time_of_generating_file
- double time_of_reading_file
- double time_of_dividing
- double time_of_sorting
- double time_of_culculating
- double time_of_writing_files

5.2.1 Function Documentation

5.2.1.1 Divide_for_two_grupse() [1/2]

```
void Divide_for_two_grupse (
            vector< Stud > & grupe,
            vector< Stud > & best_grupe,
            vector< Stud > & worst_grupe)
```

5.2.1.2 Divide_for_two_grupse() [2/2]

```
void Divide_for_two_grupse (
            vector< Stud > & grupe,
            vector< Stud > & worst_grupe)
```

5.2.1.3 Divide_for_two_grupse_v3()

```
void Divide_for_two_grupse_v3 (
            vector< Stud > & grupe,
            vector< Stud > & best_grupe,
            vector< Stud > & worst_grupe)
```

5.2.1.4 Enter_students_using_txt_file_bufer_P()

```
void Enter_students_using_txt_file_bufer_P (
            vector< Stud > & grupe)
```

5.2.1.5 Generate_file_with_students()

```
bool Generate_file_with_students (
            int number_of_students,
            int number_of_marks,
            string filename)
```

5.2.1.6 generate_marks()

```
void generate_marks (
            vector< int > & Marks,
            int number_of_marks)
```

5.2.1.7 Get_final_mark()

```
void Get_final_mark (
            vector< Stud > & grupe,
            bool for_average_homework_mark,
            bool for_both_homework_mark)
```

5.2.1.8 Get_size_for_string_printing()

```
int Get_size_for_string_printing (
            vector< Stud > & grupe)
```

5.2.1.9 Print_final_mark()

```
void Print_final_mark (
            vector< Stud > & grupe,
            bool for_average_homework_mark,
            bool for_both_homework_mark,
            bool print_results_in_terminal)
```

5.2.1.10 Sort_students()

```
void Sort_students (
            vector< Stud > & grupe,
            string parametr)
```

5.2.2 Variable Documentation

5.2.2.1 time_of_culculating

```
double time_of_culculating [extern]
```

5.2.2.2 time_of_dividing

```
double time_of_dividing [extern]
```

5.2.2.3 time_of_generating_file

```
double time_of_generating_file [extern]
```

5.2.2.4 time_of_reading_file

```
double time_of_reading_file [extern]
```

5.2.2.5 time_of_sorting

```
double time_of_sorting [extern]
```

5.2.2.6 time_of_writing_files

```
double time_of_writing_files [extern]
```

5.3 header.h

Go to the documentation of this file.

```
#pragma once

#include <iostream>
#include <iomanip>
#include <string>
#include <vector>
#include <random>
#include <fstream>
#include <sstream>
#include <chrono>
#include <algorithm>

using std::cin;
using std::cout;
using std::endl;
using std::exception;
using std::fixed;
using std::left;
using std::move;
using std::setprecision;
using std::setw;
using std::string;
using std::vector;

class Person
{
protected:
    string name_;
    string last_name_;

public:
    Person() {};
    Person(string name, string last_name) : name_(name), last_name_(last_name) {};
    inline string get_name() const { return name_; };
    inline string get_last_name() const { return last_name_; };
    virtual void about() const = 0;
    virtual ~Person() {};
};

class Stud : public Person
{
private:
    vector<int> Homework_marks_;
    int exam_mark_;
    double final_mark_ = 0;
    double second_final_mark_ = -1;

public:
    Stud() {};
    Stud(string name, string last_name) : Person(name, last_name), exam_mark_(0) {};
    Stud(std::stringstream &is, int number_of_homework_marks);
    Stud(const Stud &other) : Person(other.name_, other.last_name_),
                              Homework_marks_(other.Homework_marks_), exam_mark_(other.exam_mark_),
final_mark_(other.final_mark_), second_final_mark_(other.second_final_mark_) {};
    Stud(Stud &&other) noexcept : Person(move(other.name_), move(other.last_name_)),
                                  Homework_marks_(move(other.Homework_marks_)),
exam_mark_(other.exam_mark_), final_mark_(other.final_mark_),
second_final_mark_(other.second_final_mark_)
    {
        other.exam_mark_ = 0;
        other.final_mark_ = 0;
        other.second_final_mark_ = 0;
    };
```

```
    Stud &operator=(const Stud &other);
    Stud &operator=(Stud &&other) noexcept;
    friend std::ostream &operator«(std::ostream &out, const Stud &student);
    friend std::istream &operator»(std::istream &in, Stud &student);
    inline int get_exam_mark() const { return exam_mark_; };
    inline void set_exam_mark(int exam_mark) { exam_mark_ = exam_mark; };
    inline void set_homework_marks(int Homework_mark) { Homework_marks_.push_back(Homework_mark); };
    inline void clean_homework_marks() { Homework_marks_.clear(); };
    inline void set_final_mark(double final_mark) { final_mark_ = final_mark; };
    inline double get_final_mark() const { return final_mark_; };
    inline void set_second_final_mark(double second_final_mark) { second_final_mark_ = second_final_mark; };
    inline double get_second_final_mark() const { return second_final_mark_; };
    double Get_average_for_homework_mark();
    double Get_mediana_for_homework_mark();
    void generate_marks();
    void generate_name();
    inline void about() const override { cout « "I am a student" « endl; };
    ~Stud();
};

extern double time_of_generating_file;
extern double time_of_reading_file;
extern double time_of_dividing; // For best and worst
extern double time_of_sorting;
extern double time_of_culculating;
extern double time_of_writing_files;

void Get_final_mark(vector<Stud> &grupe, bool for_average_homework_mark, bool for_both_homework_mark);
void Print_final_mark(vector<Stud> &grupe, bool for_average_homework_mark, bool for_both_homework_mark, bool print_results_in_terminal);
int Get_size_for_string_printing(vector<Stud> &grupe);
void generate_marks(vector<int> &Marks, int number_of_marks);
void Sort_students(vector<Stud> &grupe, string parametr);
bool Generate_file_with_students(int number_of_students, int number_of_marks, string filename);
void Divide_for_two_grupse(vector<Stud> &grupe, vector<Stud> &best_grupe, vector<Stud> &worst_grupe);
void Divide_for_two_grupse(vector<Stud> &grupe, vector<Stud> &worst_grupe);
void Divide_for_two_grupse_v3(vector<Stud> &grupe, vector<Stud> &best_grupe, vector<Stud> &worst_grupe);
void Enter_students_using_txt_file_bufer_P(vector<Stud> &grupe);
```

5.4 OOP-project.cpp File Reference

```
#include "header.h"
```

Functions

- int main ()

5.4.1 Function Documentation

5.4.1.1 main()

```
int main ()
```

Index